

PRATIK RAJE

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WORK EXPERIENCE

Matrice AI (Software Developer -1)

May 2025-Present

- Reduced stream startup **latency by 67% (~15s to ~5s)** by migrating from **WebSocket to WebRTC** and implementing **trickle ICE**, buffering candidates gathered before session establishment and flushing them once the session resolved.
- **Eliminated a 4–5 second desync** between ML overlays and live video, collaborating with the ML team to match **RTP timestamps** to async inference within a ~200ms window.
- Built an adaptive canvas pipeline rendering **5–25 bounding boxes, polygons, segmentation masks and pose key-points** per frame with dynamic frame pacing and prediction reuse.
- Developed a **fault-tolerant multi-server streaming layer** running one peer connection per media server, with 5s health polling and failover after 2 consecutive failures; real-time alerting over **SSE at ~1,000 alerts/day** with **exponential-backoff reconnection**.
- Generated **typed API clients from OpenAPI contracts spanning 1,127 operations**, publishing a consumer manifest adopted by **7 producer repositories** to block breaking backend changes.
- Cut backend camera provisioning from **1000+ streams to 16 ports**, negotiating a **4x4 grid cap** with backend and product teams to remove a server-side scaling problem.
- Shipped single sign-on across two platforms via a shared NextAuth session cookie (cloud and on-premise); standardized **validation across 188 modules** with domain-organised **Zod schemas**, hardened authorization with **JWT/OAuth and CSP/HSTS**, remediating 6+ CVEs including CRLF-injection and ReDoS, with Sentry across server and edge runtimes.
- **Led frontend delivery** on the streaming platform as **top contributor across 7 repositories**, mentoring junior engineers through code review and refactoring.

PROJECTS

Support Triage Agent — Python, scikit-learn, Gemini API ·

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- Designed a 4-stage pipeline (classify → retrieve → draft → route) behind typed Pydantic contracts, each stage reading and writing to disk so any step reruns independently; swapped the entire LLM provider in ~30 lines when the original API turned out to be paid-only.
- Built the test harness before the system: a 177-case fixture, a stratified train/test split written to disk so every component scores on identical rows, and two reference implementations to beat. Final pipeline hit 65.2% precision on auto-handled traffic at 25.8% coverage, against a 36.0% baseline.
- Implemented routing as 6 ordered guard clauses ahead of the model's own confidence — billing, PII, legal, output validation — with escalation as the fall-through on every path; escalation reasons are a closed enum, so the eval reports which guard fired how often rather than an undifferentiated rate.
- Fixed three silent correctness bugs found by cloning and running the repo fresh: a test-set leak into the search index, a CRLF-on-checkout bug invalidating 408 cached fixtures, and a harness reporting an accuracy figure for API calls that had all failed. Every model response is cached by hash, so headline numbers reproduce in 30 seconds with no API key. 48 documented ADRs.

MCU Hub- Next.js, Express, MongoDB

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- Implemented a graph traversal service over 193 nodes / 755 edges (BFS, weighted Dijkstra, centrality), cutting shortest-path p95 from 34.7ms to 0.2ms with an in-process adjacency snapshot invalidated on write.
- Designed a REST API across 8 domain modules (vertical slices owning model, service, controller, routes and validators) with Zod validation, cursor pagination, rate limiting and centralized error handling.
- Built JWT auth with refresh-token rotation and reuse detection (15-minute in-memory access tokens, versioned 7-day refresh tokens in httpOnly cookies), plus idempotency keys on writes returning 409 in-flight and 422 on key reuse with a different payload, so a client retry after a dropped connection cannot double-write.
- 41 tests verified by mutation testing; route-layer tests against in-memory MongoDB caught two production 500s that should have been 400s. Deployed on Render with MongoDB Atlas; CI runs lint, typecheck and tests on Node 20 and 22 per PR, with 5 documented architecture decisions including why not Neo4j.

SKILLS

- **Frontend:** React, Next.js, TypeScript, JavaScript, Redux Toolkit, Tailwind CSS, Material UI, HTML5, CSS3
- **Real-Time & Video:** WebRTC, WebSockets, Server-Sent Events, RTP, Canvas API
- **Backend & Databases:** Node.js, Express.js, MongoDB, Mongoose, OpenAPI, REST API design
- **Python & ML:** Python, Pydantic, scikit-learn, pandas, LLM pipelines, evaluation design
- **Auth & Security:** NextAuth, JWT/OAuth, Zod, CSP/HSTS, CVE remediation, Sentry
- **Testing & Tooling:** Vitest, Supertest, mongodb-memory-server, GitHub Actions (CI/CD), Git, Postman, Vercel

EDUCATION

Indian Institute of Information Technology, Bhopal

December 2021 - June 2025

Bachelor of Technology in Information Technology (IT)

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Computer Networks, Operating Systems